POWER BI IMPLEMENTATION & DAX MODELING GUIDE

Step-by-Step Architecture to Replicate the Business Operations Dashboard in Power BI Desktop

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Important Technical Note: *Standard server execution environments cannot programmatically compile proprietary Microsoft Power BI binary archives (.pbix). As a Senior Business Operations Analyst and Mentor, this comprehensive implementation guide provides the full dimensional architecture, data types, exact DAX formula code, visual field mappings, and cross-reconciliation tables required to assemble the production report in Power BI Desktop in under 15 minutes.*

1. Data Ingestion & Power Query Transformation

Step 1: Ingestion
• Launch Power BI Desktop → Select **Get Data** → **Text/CSV**.
• Import the generated Business_Operations_Clean_Data.csv.

Step 2: Column Data Type Verification in Power Query
• Record ID: Text
• Month: Date (or Text formatted as YYYY-MM)
• Department: Text
• Employee Count: Whole Number (Int64)
• Working Days: Whole Number (Int64)
• Target Units: Whole Number (Int64)
• Actual Units: Whole Number (Int64)
• Delayed Tasks: Whole Number (Int64)
• Operating Cost (INR): Fixed Decimal Number / Currency (INR)
• Customer Complaints: Whole Number (Int64)
• Click **Close & Apply**.

2. Data Modeling & Star Schema Architecture

While the analysis can be driven directly from the single flat fact table (Business_Operations_Clean_Data), establishing a standard Star Schema elevates recruiter evaluation:
• **Fact Table:** Fact_Operations (from Clean Data)
• **Dimension 1 (Date):** Dim_Date linked via Fact_Operations[Month] (1-to-Many, Single Direction)
• **Dimension 2 (Department):** Dim_Department linked via Fact_Operations[Department] (1-to-Many)

3. Production-Ready DAX Measures

Create a dedicated measure table named _KeyMeasures and insert the following tested DAX expressions:

Measure Name	DAX Expression	Format / Category
Total Target Units	Total Target Units = SUM(Fact_Operations[Target Units])	Whole Number (#,##0)
Total Actual Units	Total Actual Units = SUM(Fact_Operations[Actual Units])	Whole Number (#,##0)
Target Achievement %	Target Achievement % = DIVIDE([Total Actual Units], [Total Target Units], 0)	Percentage (0.00%)
Total Delayed Tasks	Total Delayed Tasks = SUM(Fact_Operations[Delayed Tasks])	Whole Number (#,##0)
Delay Rate %	Delay Rate % = DIVIDE([Total Delayed Tasks], [Total Actual Units], 0)	Percentage (0.00%)
Total Operating Cost	Total Operating Cost = SUM(Fact_Operations[Operating Cost (INR)])	Currency (■#,##0)
Cost per Completed Unit	Cost per Completed Unit = DIVIDE([Total Operating Cost], [Total Actual Units], 0)	Currency (■#,##0.00)
Total Complaints	Total Complaints = SUM(Fact_Operations[Customer Complaints])	Whole Number (#,##0)
MoM Actual Growth %	MoM Actual Growth % = VAR PrevMonth = CALCULATE([Total Actual Units], DATEADD(Dim_Date[Date], -1, MONTH)) RETURN DIVIDE([Total Actual Units] - PrevMonth, PrevMonth, BLANK())	Percentage (0.00%)

4. Visual Field Configurations (Canvas Setup)

Visual 1: KPI Cards (Top Banner)
• Add 5 'Card (new)' visuals across top row: [Total Target Units], [Total Actual Units], [Target Achievement %], [Total Operating Cost], [Cost per Completed Unit].

Visual 2: Monthly Target vs Actual (Line and Clustered Column Chart)
• X-axis: Month | Column Y-axis: Total Target Units, Total Actual Units | Line Y-axis: Target Achievement %.

Visual 3: Department Performance Matrix (Clustered Bar Chart)
• Y-axis: Department | X-axis: Delay Rate % | Tooltips: Total Delayed Tasks, Customer Complaints.

Visual 4: Operating Cost Escalation Trend (Area Chart)
• X-axis: Month | Y-axis: Total Operating Cost | Legend: Department.

Visual 5: Slicers & Interactivity
• Add horizontal chiclet slicer for Department.
• Add dropdown slicer for Month range.

5. Excel-to-Power BI Data Reconciliation Checklist

Before presenting to recruiters, cross-verify that the DAX card outputs match the Excel ground truth:
• **Total Actual Units:** Must equal exactly **74,164**.
• **Overall Target Achievement %:** Must equal exactly **95.90%**.
• **Total Operating Cost:** Must equal exactly ■**15,789,176**.
• **Total Customer Complaints:** Must equal exactly **1,974**.